After Excel

We can:

- Clean data
- Calculate totals
- Create dashboards

➔ One question→ How do we know whether the number we calculated tells us something useful

**Company→100,000 → 0.1 million**

What is average amount of our customers spend every month, total profit, average profit, profit margin, year over year growth?

**But now questions are different?**

- What if there are 10 million customers?
- What if customers are spread across multiple regions?
- What if collecting the data is expensive?
- What if we need answer quickly?

➔ The conditions create a need for statistics

"Statistics helps us make reliable conclusions from the data, especially when we cannot practically study everything."

# What is Statistics?

**Statistics is the science of collecting, organizing, analysing, interpreting and presenting data to make better decisions.**

## 1. Collect: Get the required data

Example: A company collects

- Customer age
- Location
- Orders
- Sales
- Ratings

## 2. Organize: Arrange the data in a useful format (table structure)

| Customer_Name | Region | Phone | Orders | Price |
|---|---|---|---|---|
| ABC | India | 1234567890 | 10 | $12,345 |

## 3. Analyze: Calculate and investigate

Example:

- Average (Mean) Sales
- Median Spending
- Sales Variation
- Customer distribution

## 4. Interpret

This is where analysts create a meaning.
Suppose: AVG monthly customer spending = **$4,500**

➔ This calculation is not an insight
The analyst asks:
Spending on What? → Category of spending→ Customer behaviour( electrical, furniture, cosmetics)
100 people watch a movie trailer→ 90 people finds interesting→50 of them have budget →40 might watch the movie

## 5. Present: Communicate the result

- Dashboard

➔ Chart
➔ Report
➔ Presentation
➔ Business recommendation

**Statistics is not just formulas**

➔ Statistics = Mean + Median + Standard Deviation (X)

➔ It is a process

- **DATA→ ANALYSIS→ INTERPRETATION→ (PRESENT)→ DECISION**

**5,000,000 orders**

Collect→ customer transaction data
Organize→ Clean and structure the data
Analyze→ Calculate average spending, total revenue, average profit margin, total profit
Interpret→ Understand customer spending behaviour
Present→ Create a report/dashboard
Decision→ Design customer segments or offers

If we are increasing the discount→ It will attract the customers
MRP→9,999→ 3,999 (after discount) 10k→ 4k (60% discount)
Increase discount same product
MRP→ 11,999→ 2,999→ 2,499 12k→ 2.5k (80% discount)

**Where is statistics used?**

- **Business analytics**
- **Finance**
- **Operations**
- **HR analytics**
- **Data Science**
- **Marketing**

**If I calculate the average sales of 500 transactions, have I done the statistics?**

Not necessarily

AVG calculation is one single operation

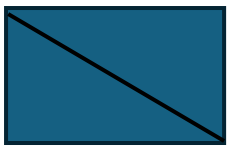
Statistics also involves understanding:

- What data was collected?
- Is it representative? 100 txn AVG 500 total price 50000
- What does the result mean?
- Can we generalize it?
- What decision should we make?

## Importance of Statistics in  Data Analytics

- Converts raw data into meaningful insights.
- Helps identify trends, patterns and relationships
- Support data driven decisions
- Helps forecast outcomes
- Reduce uncertainity
- Supports Data Science models
- Is important for A/B testing, surveys and experiments.

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**1. Converts Raw data into insghts**

₹10,000

₹12,000

₹8,000

₹15,000

₹11,000

Statistics help us :

- Average, SUM
- Total values( number of transactions)
- Variance in sales
- Unusual values

**2. Identify trends and patterns**

**Company monthly sales**

| **Month** | **Sales** |
|---|---|
| Jan | ₹10L |
| Feb | ₹11L |
| Mar | ₹12L |
| Apr | ₹15L |
| May | ₹16L |

The analyst can identify upward pattern( growth)

➔ Next months will also have higher sales with respect to previous months

- **A pattern in historical data doesn't automatically guarantee the future.**

## 3. Support Data Driven Decisions

→ Increasing the budget of advertisement

Instead of :

" I think advertising is working"

Analyst thinking:

What does the data show?, Is Product in demand?

➔ Evidence based statistics.

**** 4. Reduce Uncertainty**

**Statistics does not remove / eliminate uncertainty**

**It helps us measure and manage uncertainty.**

Example:

A company wants to estimate customer satisfaction

➔ They don't know opinion of each customer
➔ 10000→ 2 people are conducting survey→ 5 days→ Sampling(randomly pick 1000 customers)
➔ Random Sampling (un-biased)

## 5. Data Science:

Statistics provides the foundation for many Data Science Techniques.
Example:

- Probability
- Distribution
- Sampling
- Regression
- Model Evaluation

**Machine learning can give you a prediction, but Statistics helps you understand the data, uncertainty, relationships and evidence behind that analysis.**

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## Applications in Real Life Business:

- **Sales & Marketing**
  - Sales trends
  - Customer behaviour
  - Campaign performance

1st 5lakh → 12lakh
2nd 5 lakh → 15 lakh
Statistics evaluate the evidence.

- **Finance**
- `Risk analysis
- Investment returns
- Fraud Detection

Example: A bank analyses the txn behavious

If a txn is very unusual with normal customer behaviour, it may require investigation.

- **Operations**
- Process improvement
- Quality control
- Demand forecasting

Example: A company records defects from production batches
Statistics helps in determine
" Is the defect rate stable, or is something wrong with the production process?"

- **HR Analytics**
- Employee performance
- Attrition analysis
- Salary benchmarking
  Example: HR wants to know:
  Do employees with more experience earns significantly higher salaries.
  Correlation, regression or hypothesis testing

- **DATA Science:**
- Model building
- Prediction
- Machine learning
- AI
  "Statistics is not the whole of Data Science but it is one of the important foundation."

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**Population: The entire group of individuals or items that we are interested in studying.**

**Sample: A subset or a part of the population selected for study**

A class of students→ Population (100)
The students with hight more than 160cms from that class→ Sample (40)

**Why don't we always study the population?**

**1. Cost**

**2. Time**

**3. Accessibility**

**4. Practicality:**

➔ **The purpose of sampling is not merely to collect less data**

**Main Goal:** **Collect enough information data to make reliable conclusions about the population**

**Population vs Sample**

| **Population** | **Sample** |
|---|---|
| Entire group | Part of the group |
| Usually larger | Usually smaller |
| Target study | Used to study the target |
| Can be expensive to measure | More practical |
| Example: All customers (10,000) | Example: 10% of customers (1,000) |
| | |